• **Inference (d) is also not correct,** because the passage does not discusses that "with high density of population and widespread poverty in our urban areas, planned development in them is very difficult. It is clear that density of population and poverty make development planning is not specified in the passage.

22. Based on the above passage, the following assumptions have been made:
1. India needs a new generation of urban professionals with knowledge relevant to modern urban practice.
2. Indian universities at present have no capacity or potential to impart training in systems approach.
Which of the assumptions given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans.(a): Assumption 1 is valid, because the statement 'India needs a new generation of Urban professionals with knowledge relevant to modern urban practice, is relevant with the passage. The passage discusses that there is a need for professional trained in modern system approaches to urban planning.
• **Assumption 2 is not valid,** because the statement 'India universities at present have no capacity or potential to import training in system approach, is not appropriate according to the passage.

Passage–2

Not every voice on the internet commands the same kind of audience. When anonymous private entities with high capital can pay for more space for their opinions, they are effectively buying a louder voice. If political discourse in the digital sphere is a matter of outshining one's opponents till the election is won, then the quality of politics suffers. The focus of social media is restricted to the promotion of content that generates more user engagement, regardless of how inflammatory the content may be.

23. Which one of the following statements best reflects the central idea of the above passage?
(a) Constructed as marketplace of views, social media ensures instant access to information
(b) Social media are not ideal or moral institutions but the products built by companies to make profits
(c) Social media have been created to strengthen democracies
(d) In today's world social media are inevitable for well-informed social life.

Ans. (b) : Options (a) is not correct, because the passage does' not focus on instant access to information, but rather on the influence of capital in amplifying voices. The statement "Constructed as a market place of views, social media ensures instant access to information," does not make sure the social media is only form to access information.
• **Options (b) is correct,** because the statement "Social media are not ideal or moral institutions but the products built by companies to make profit is appropriate according to the passage. The passage focuses the social media platforms are driven by profit motives, leading to prioritization of user engagement and inflammatory content.
Options (c) is not correct, because the social media platforms are not created only to strengthen political scenario and democracy. If may contradict the ideal of strengthening democracies.
Options (d) is also not correct, because the passage does not discuss that social media are inevitable for well-informed social life. The social media platform has many flaws and negative impact. It may create political chaos.

24. Based on the above passage, the following assumptions have been made:
1. Internet is not inclusive enough.
2. Internet can adversely affect the equality of politics in a country.
Which of the assumptions given above is/are valid?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : Assumption 1 is valid, because the passage points out that in political scenario, the intent is not inclusive enough. There are multiple social media platforms, to be the part of strengthening the idea of democracy.
• **Assumption 2 is also valid,** because the passage clearly discusses the internet con adversely affect the quality of politics in a country.

25. $222^{333} + 333^{222}$ is divisible by which of the following numbers?
(a) 2 and 3 but not 37 (b) 3 and 37 but not 2
(c) 2 and 37 but not 3 (d) 2, 3 and 37

Ans. (b) : $\Rightarrow 222^{333} + 333^{222}$
$= (2 \times 111)^{333} + (3 \times 111)^{222}$
$= 111^{222} \times (2^{333} \times 111^{111} + 3^{222})$
As, $111 = 3 \times 37$
So, the number is divisible by 3 and 37.
When we divide number by 2,
$2^{333} \times 111^{111}$ is divisible by 3^{222} is an odd number and not divisible. So, the number is an odd number and not divisible by 2.

26. What percent of water must be mixed with honey so as to gain 20% by selling the mixture at the cost price of honey?
(a) 20% (b) 10% (c) 5% (d) 4%

Ans. (a) : Let amount of honey is = H unit
Water added = W unit
according to question–

$$\text{Profit} = \left(\frac{\text{Selling price} - \text{cost price}}{\text{cost price}}\right) \times 100\% = 20\%$$

Where the price of honey is H.

$$\frac{H + W - H}{H} = \frac{20}{100} = 0.2$$

$$\frac{W}{H} = 0.2$$

$$W = 0.2H$$

So, water should be added 20% of honey to gain profit of 20%.

27. What is the rightmost digit preceding the zeros in the value of 30^{30}?
(a) 1 (b) 3 (c) 7 (d) 9

Ans. (d) : Right most digit preceding the zeros of 30^{30}.
$\Rightarrow 30^{30} = 30^{30} \times 10^{30}$
for right most digit preceding zero we have to take 3^{30} and find out unit digit of 3^{30}.
$\Rightarrow$ We have to divide power of 3 by 4 and find out its reminder.

$$\frac{30}{4} \rightarrow 2 \text{ (Reminder)}$$

So, right most unit digit would be $3^2 = 9$

28. 421 and 427, when divided by the same number, leave the same remainder 1. How many numbers can be used as the divisor in order to get the same remainder 1?
(a) 1 (b) 2
(c) 3 (d) 4

Ans. (c) : $\Rightarrow 421 - 1 = 420 = 2^2 \times 3^1 \times 5^1 \times 7^1$
$427 - 1 = 426 = 2^1 \times 3^1 \times 7^1$
Common factors of 420 & 426 = 2, 3
Dividing by 2 & 3 Reminder would be $\perp$ from number 2 & 3, 6 also would be a number by which reminder will be $\perp$. So total number by which we get reminder $\perp$ after dividing, are 2, 3 & 6.

29. A can X contains 399 litres of petrol and a can Y contains 532 litres diesel. They are to be bottle in bottles of equal size so that whole of pertorl and diesel would be separately bottle. The bottle capacity in terms of litres is an integer. How many different bottle sizes are possible?
(a) 3 (b) 4 (c) 5 (d) 6

Ans. (b) : Common factors of 399 & 532,

$$\left.\begin{aligned} 399 &= 3^1 \times 7^1 \times 19^1 \times 133 \\ 532 &= 2^2 \times 7^1 \times 19^1 \times 133 \end{aligned}\right\} \text{common factors} = 1, 7, 19, 133$$

So, there can be 4 type of bottle having capacity 1, 7, 19, & 133 litres.

30. Consider the following statements in respect of the sum S = x + y + z, where x, y and z are distinct prime numbers each less than 10:
1. The unit digit of S can be 0.
2. The unit digit of S can 9.
3. The unit digit of S can be 5.
Which of the statement given above are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (c) : S = x + y + z
Where x, y, and z are distinct prime number each less than 10,
Prime number less than 10 = 2, 3, 5, 7
Let's assume x = 2, y = 3, z = 5
Then S = 2 + 3 + 5 = 10 $\Rightarrow$ Unit digit is 0.
Again take x = 3, y = 5, z = 7
Then S = 3 + 5 + 7 = 15 $\Rightarrow$ Unit digit is 5.
But there is no any condition by which we can get 9 at unit digit.
Hence statement (1) and (3) are correct.

Directions for the following 4 (four) items:
Read the following two passages and answer the items that follow the passages. Your answer to these items should be based on the passages only.

Passage–1

According to the Food and Agriculture Organization, one-third of food produced for human consumption is lost or wasted globally. Food is lost or wasted throughout the supply Food is lost or wasted throughout the supply chain, from initial agricultural production to final household consumption. The increasing wastage also results in land degradation by about 45%, mainly due to deforestation, unsustainable agricultural practices, and excessive groundwater extraction. The energy spent over wasted food results in about 3.5 billion tonnes of carbon dioxide production every year. Decay also leads to harmful emissions of other gases in the atmosphere. Addressing the loss and wastage of food in all forms is critical to complete the cycle of food sufficiency and food sustainability.

31. Which of the following statements best reflect the most logical and rational inferences that can be made from the passage?
1. The current methods of food distribution are solely responsible for the loss and wastage of food.
2. Land productivity is adversely affected by the prevailing trend of food loss and wastage.
3. Reduction in the loss and wastage of food results in lesser carbon footprint.
4. Post-harvest technologies to prevent or reduce the loss and wastage of food are not available.
Select the correct answer using the code given below.
(a) 1, 2 and 3 (b) 2 and 3 only
(c) 1, 3 and 4 (d) 1, 2 and 4

Ans. (b) : Inference 1 is not correct- This statement is according to the passage this suggest that food loss and wasted throughout the supply chain from production to household consumption, and not solely due to distribution methods.
• **Inference 2 is correct.** The land productivity is adversely affected by the prevailing trend of food loss and wastage. This means that wastage increases land degradation, which adversely affects land productivity.
• **Inference 3 is correct.** Reduction in the loss and wastage of food result in lesser carbon foot print. This statement is appropriate according to the passages, which mention that the energy spent on wasted food results in significant carbon dioxide production. Thus, the reaction in food loss and waste would reduce the carbon footprint.
• **Inference 4 is not correct.** Post harvest technologies to prevent or reduce the loss and wattage of food according to the passage. This passage does not defines the non-availability of post-harvest technologies. It is know to all that various technologies and methods are exist to reduce food loss and waste after harvest.

32. Based on the above passage, the following assumptions have been made:
1. The food distribution mechanism needs to be reimagined and made effective to reduce the loss and wastage of food.
2. Ensuring the reduction of wastage and loss of food is a social and moral responsibility of all citizens.
Which of the assumptions given below is/are valid?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (b/c) : Assumption 1 is valid. The food distribution mechanism needs to be reimagined and made effective to reduce the loss and wastage of food. This passage discusses that food is lost or wasted throughout the supply chain. While it does not discuss only about the distribution mechanism, it implies that improvements across the entire supply chain, including distribution, would help to reduce the food wastage. So, assumption is appropriate and valid.
• **Assumption is not valid.** This passage discusses about the environmental and economic aspects of food wastage and need address it to achieve food efficiency and sustainability.
Note – In this passage we find that the entire passage focuses on the loss and wastage of food and its impact on the environment and humans. We also find that the passage does explicitly connect the social and moral duty of citizens to reduce wastage and loss of food. So, this passage does not indicate a moral and social responsibility on citizens but, they too must ensure the reduction of wastage and loss of food.